

Proposition 2.4. *There is a bijection between the k -dimensional faces of P that contain $\mathbf{v}$, and the $(k-1)$ -dimensional faces of $P/\mathbf{v}$, given by*

$$\begin{aligned}\pi : F &\longmapsto F \cap \{\mathbf{x} : \mathbf{c}\mathbf{x} = c_1\}, \\ \sigma : P \cap \text{aff}(\{\mathbf{v}\} \cup F') &\longleftarrow F'.\end{aligned}$$

Proof. Denote the “cutting hyperplane” by $H := \{\mathbf{x} : \mathbf{c}\mathbf{x} = c_1\}$.

The map π is well defined: let $F = P \cap \{\mathbf{x} : \mathbf{b}\mathbf{x} = b_0\}$; then we get $F \cap H = (P \cap H) \cap \{\mathbf{x} : \mathbf{b}\mathbf{x} = b_0\}$, where $\mathbf{b}\mathbf{x} \leq b_0$ is valid for P and thus also for $P/\mathbf{v}$.

To see that σ is well defined, let $F' = (P \cap H) \cap \{\mathbf{x} : \mathbf{b}\mathbf{x} = b_0\}$, where $\mathbf{b}\mathbf{x} \leq b_0$ is valid for $P/\mathbf{v}$. Then $(\mathbf{b} + \lambda\mathbf{c})\mathbf{x} \leq b_0 + \lambda c_1$ is valid for $P/\mathbf{v}$, for all $\lambda \in \mathbb{R}$. Now a simple computation shows that for

$$\lambda_0 := \frac{b_0 - \mathbf{b}\mathbf{v}}{c_0 - c_1},$$

the inequality is valid on P , with equality for $\mathbf{v}$. In fact, if we consider $\mathbf{v}' \in V \setminus \mathbf{v}$, then we know $\mathbf{c}\mathbf{v}' < c_1$ and $\mathbf{c}\mathbf{v} = c_0 > c_1$, so

$$\mathbf{v}'' := \frac{(\mathbf{c}\mathbf{v} - c_1)\mathbf{v}' + (c_1 - \mathbf{c}\mathbf{v}')\mathbf{v}}{\mathbf{c}\mathbf{v} - \mathbf{c}\mathbf{v}'} \in P \cap H = P/\mathbf{v}.$$

Now $\mathbf{v}''$ is a convex combination of $\mathbf{v}$ and $\mathbf{v}'$, and from $(\mathbf{b} + \lambda_0\mathbf{c})\mathbf{v}'' \leq b_0 + \lambda_0 c_1$ and $(\mathbf{b} + \lambda_0\mathbf{c})\mathbf{v} = b_0 + \lambda_0 c_1$ we get $(\mathbf{b} + \lambda_0\mathbf{c})\mathbf{v}' \leq b_0 + \lambda_0 c_1$.

Now we check that the maps σ and π are inverses of each other: we compute

$$\pi \circ \sigma(F') = H \cap P \cap \text{aff}(\{\mathbf{v}\} \cup F') = P \cap H \cap \text{aff}(F') = P/\mathbf{v} \cap \text{aff}(F') = F',$$

where the last equality is from Proposition 2.3(iv). Similarly,

$$\sigma \circ \pi(F) = P \cap \text{aff}(\{\mathbf{v}\} \cup (F \cap H)) \stackrel{*}{=} P \cap \text{aff}(F) = F,$$

where for the equality $*$ we use that every vertex $\mathbf{v}'$ of F can be obtained as an affine combination of $\mathbf{v}$ and a point $\mathbf{v}'' \in F \cap H$.

Finally we observe that if F' is a face of $P/\mathbf{v}$ of dimension $k-1$, then the associated face F of P has dimension k , since its affine hull is $\text{aff}(F) = \text{aff}(F' \cup \{\mathbf{v}\})$, where $\mathbf{v} \notin \text{aff}(F')$ by construction. $\square$

2.2 The Face Lattice

In this section we translate some of our results on polytopes into purely combinatorial statements. For this we need some terminology about finite *partially ordered sets* (“posets,” for short). We refer to Stanley’s book [517, Sects. 3.1–3.3] for more information on that subject and its ramifications. For simplicity, and to unify terminology, we define the key concepts here.

Definition 2.5 (Poset terminology).

A *poset* $(S, \leq)$ is a finite partially ordered set, that is, a finite set S equipped with a relation “ $\leq$ ” which is reflexive ($x \leq x$ for all $x \in S$), transitive ($x \leq y$ and $y \leq z$ imply $x \leq z$), and antisymmetric ($x \leq y$ and $y \leq x$ imply $x = y$).

Usually we denote such a poset by S , when the partial order is clear. Any subset of S is also a poset, with the induced partial order. A *chain* in S is a totally ordered subset of S ; its *length* is its number of elements minus 1.

For elements $x, y \in S$ with $x \leq y$, we denote by

$$[x, y] := \{w \in S : x \leq w \leq y\}$$

the *interval* between x and y . An interval in S is *boolean* if it is isomorphic to the poset $B_k = (2^{[k]}, \subseteq)$ of all subsets of a k -element set, for some k .

A poset is *bounded* if it has a unique *minimal* element, denoted $\hat{0}$, and a unique *maximal* element, denoted $\hat{1}$. The *proper part* of a bounded poset S is $\bar{S} := S \setminus \{\hat{0}, \hat{1}\}$.

A poset is *graded* if it is bounded, and every maximal chain has the same length. In this case the length of a maximal chain in the interval $[\hat{0}, x]$ is the *rank* of x , denoted by $r(x)$. The rank $r(S) := r(\hat{1})$ is also called the *length* of S . For example, every chain is a graded poset, with $r(C) = |C| - 1$, and the boolean posets B_k are graded of length $r(B_k) = k$, for all $k \geq -1$.

A poset is a *lattice* if it is bounded, and every two elements $x, y \in S$ have a unique minimal upper bound in S , called the *join* $x \vee y$, and every two elements $x, y \in S$ have a unique maximal lower bound in S , called the *meet* $x \wedge y$. (In fact, any two of these three conditions imply the third; also, if every pair of elements has a join respectively meet, then also every finite subset has a join respectively meet.)

If S is a graded lattice, then we call the minimal elements of $S \setminus \hat{0}$ its *atoms*, and the maximal elements of $S \setminus \hat{1}$ its *coatoms*. Equivalently, the atoms are the elements of rank 1, and the coatoms are the elements of rank $r(S) - 1$.

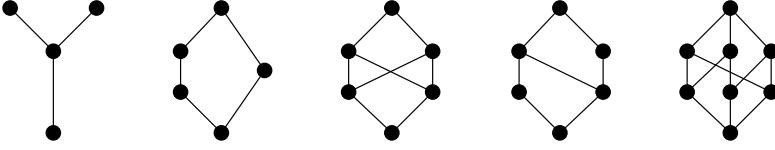
A lattice is *atomic* if every element is a join $x = a_1 \vee \dots \vee a_k$ of $k \geq 0$ of atoms, where we obtain $x = \hat{0}$ for $k = 0$, and an atom $x = a_1$ for $k = 1$. Similarly, a lattice is *coatomic* if every element is a meet of coatoms.

We define the *opposite poset* S^{op} (or *order dual*) to have the same underlying set as S , with $x \leq y$ in S^{op} if and only if $y \leq x$ holds in S .

We use the graphical representation of posets by *Hasse diagrams*, that is, graphs drawn in the plane so that the elements correspond to vertices, where $x \leq y$ holds if and only if there is an increasing path from x to y . Here we only include the edges corresponding to *cover relations*, that is, if $x < y$ and $[x, y] = \{x, y\}$.

Of the posets in the following figure, the first one is not bounded, but all others are. The second one is a lattice, but not graded. The third one

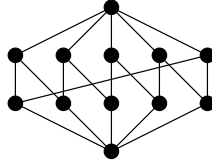
is graded (of length 3), but it is not a lattice. The fourth poset is a graded lattice (of length 3), and the fifth one is even boolean (isomorphic to B_3). The fourth poset is neither atomic nor coatomic, but the fifth one is.



Why is all this interesting for us? Because we want to study the set of faces of a convex polytope, ordered by inclusion.

Definition 2.6. The *face lattice* of a convex polytope P is the poset $L := L(P)$ of all faces of P , partially ordered by inclusion.

The following figure shows, as an example, the face lattice of a convex pentagon. Here the minimal element corresponds to the empty face, the five atoms in the layer above correspond to the five vertices, the layer above this represents the five edges (each containing two vertices), and the top element represents the pentagon itself.



In Theorem 2.7 we collect the main structural properties of face lattices, starting with the fact that they *are* lattices, justifying the terminology of Definition 2.6.

Theorem 2.7. Let P be a convex polytope.

- (i) For every polytope P the face poset $L(P)$ is a graded lattice of length $\dim(P) + 1$, with rank function $r(F) = \dim(F) + 1$.
- (ii) Every interval $[G, F]$ of $L(P)$ is the face lattice of a convex polytope of dimension $r(F) - r(G) - 1$.
- (iii) (“Diamond property”) Every interval of length 2 has exactly four elements. That is, if $G \subseteq F$ with $r(F) - r(G) = 2$, then there are exactly two faces H with $G \subset H \subset F$, and the interval $[G, F]$ looks



- (iv) The opposite poset $L(P)^{op}$ is also the face poset of a convex polytope.
- (v) The face lattice $L(P)$ is both atomic and coatomic.

Proof. To see that $L(P)$ is a lattice it suffices to see that it has a unique maximal element $\hat{1} = P$ and a unique minimal element $\hat{0} = \emptyset$, and that meets exist, with $F \wedge G = F \cap G$; this is true because $F \cap G$ is a face of F and of G , and thus of P , by Proposition 2.3(ii). And clearly every face of P that is contained in F and in G must be contained in $F \cap G$.

We continue with part (ii). For this we can assume that $F = P$, by Proposition 2.3(iii). Now if $G = \emptyset$, then everything is clear. If $G \neq \emptyset$, then it has a vertex $\mathbf{v} \in G$ by Proposition 2.2(i), which is a vertex of P by Proposition 2.3(iii). Now the face lattice of $P/\mathbf{v}$ is isomorphic to the interval $[\{\mathbf{v}\}, P]$ of the face lattice $L(P)$, by Proposition 2.4. Thus we are done by induction on $\dim(G)$.

For part (i) it remains to see that the lattice $L(P)$ is graded. If $G \subset F$ are faces of P , then from $G = P \cap \text{aff}(G) \subseteq P \cap \text{aff}(F) = F$, which holds by Proposition 2.3(iv), we can conclude that $\text{aff}(G) \subset \text{aff}(F)$, and thus that $\dim(G) < \dim(F)$. So it suffices to show that if $\dim(F) - \dim(G) \geq 2$, then there is a face $H \in L(P)$ with $G \subset H \subset F$. But by part (ii) the interval $[G, F]$ is the face lattice of a polytope of dimension at least 1, so it has a vertex, which yields the desired H .

Part (iii) is a special case of (ii): the “diamond” is the face lattice of a 1-dimensional polytope.

We don’t prove part (iv) here — but we will do so in the next section.

Finally, for part (v), the first part is immediate from Proposition 2.2(i), where the atoms of $L(P)$ correspond to the vertices of P , and the second part follows from this by taking the opposite poset, according to part (iv). $\square$

This theorem contains quite restrictive information on the structure of polytope face lattices (Exercise 2.3). We will get even more precise information later.

We note here that the face lattice is the proper framework to define combinatorial equivalence of polytopes. In fact, our previous definition (before Example 0.2) can be restated as saying that P and Q are *combinatorially equivalent*, $P \simeq Q$, if and only if $L(P) \cong L(Q)$: if their face lattices are isomorphic.

By Proposition 2.2(i), this is equivalent to a bijection $\text{vert}(P) \leftrightarrow \text{vert}(Q)$ between the vertices of P and Q , in such a way that the vertex sets of faces of P correspond (under this bijection) to the vertex sets of faces of Q . A general observation is that in this context it is enough to deal with vertices and facets, because the faces are exactly the intersections of facets, and the vertex sets of faces are exactly the intersections of vertex sets of facets — see Exercise 2.7. (Abstractly, the key properties are that face lattices are atomic and coatomic.)

Topologically, combinatorial equivalence corresponds to the existence of a (piecewise linear) homeomorphism between the polytopes $P \cong Q$ that restricts to homeomorphisms between the facets (and hence all the faces)

of P and Q . In other words, $P \cong Q$ if and only if P and Q define isomorphic cell complexes (CW-balls) — see Munkres [418] or Björner [89, Sect. 12] [96, Sect. 4.7] for these concepts.

2.3 Polarity

We proceed to construct *polar polytopes*: this is what (nearly) everybody else calls the “dual” of a polytope. We will use the term “polar” in order to distinguish polarity from duality in the sense of (oriented) matroid theory, which we will see later in Lecture 6 in the form of Gale diagrams, the Lawrence construction, and others. (In this, we follow the conventions of [96, pp. 44–45].)

A key observation one should not miss is the *step into dual space* we take in this section. Equivalently, one could just fix a scalar product on $\mathbb{R}^d$, and this way we could construct a polar polytope in the same space as the original. However, there is a lot of choice in any case because the location of the origin is essential for our construction. If we wanted to avoid this, we would have to linearize, and develop “cone polarity” — using some methods of Lecture 1. We will not do this here (in order to keep the principle of making this a lecture on polytopes), but see Exercise 2.13.

Lemma 2.8. *Let P be a polytope in $\mathbb{R}^d$. Then the following conditions are equivalent for $\mathbf{y} \in P$.*

- (i) $\mathbf{y}$ is not contained in a face of P of dimension smaller than d ,
- (ii) if $\mathbf{a}\mathbf{y} = a_0$ and $\mathbf{a} \neq 0$, then $\mathbf{a}\mathbf{x} \leq a_0$ is not valid for P ,
- (iii) $\mathbf{y}$ can be represented in the form $\mathbf{y} = \sum_{i=0}^d \lambda_i \mathbf{x}_i$ for $d+1$ affinely independent points $\mathbf{x}_0, \dots, \mathbf{x}_d \in P$ and for parameters $\lambda_i > 0$ with $\sum_{i=0}^d \lambda_i = 1$,
- (iv) $\mathbf{y}$ can be represented as $\mathbf{y} = \frac{1}{d+1} \sum_{i=0}^d \mathbf{x}_i$ for $d+1$ affinely independent points $\mathbf{x}_0, \dots, \mathbf{x}_d \in P$.

Proof. Part (i) implies that P is full-dimensional. From this we get that part (ii) holds: if $\mathbf{a}\mathbf{x} \leq a_0$ were valid for P , then $\mathbf{y}$ would be contained in the face $P \cap \{\mathbf{x} \in \mathbb{R}^d : \mathbf{a}\mathbf{x} = a_0\}$ of smaller dimension. Conversely, if part (ii) holds, then no inequality can define a facet that contains $\mathbf{y}$.

Part (iv) trivially implies part (iii), and from this we get part (ii) by an easy calculation:

$$a_0 = \mathbf{a}\mathbf{y} = \sum_{i=0}^d \lambda_i \mathbf{a}\mathbf{x}_i \leq \sum_{i=0}^d \lambda_i a_0 = a_0.$$